AQ9.1: Activity Questions 1 - Not Graded

This assignment will not be graded and is only for practice.
Functions of several variables:

Level 1:

1 point
Which of the following are correct with respect to $f(x, y) = x^2 - 2y$, where x, y are real?

- [ ] Domain of f is R^2
- [ ] Domain of f is R
- [ ] Range of f is R
- [ ] Range of f is R^2

No, the answer is incorrect.
Score: 0
Accepted Answers:
Domain of f is R^2
Range of f is R

1 point
Let $D \subset R^n$ and $f: D \longrightarrow R^m$, $g: D \longrightarrow R^m$ be multivariable functions on D. Which of the following statements is(are) TRUE?

- [ ] The product function fg is defined on D by $fg(\tilde{x}) = f(\tilde{x}) \times g(\tilde{x}), \tilde{x} \in D$
- [ ] The sum function $f + g$ is defined on D by $(f + g)(\tilde{x}) = f(\tilde{x}) + g(\tilde{x}), \tilde{x} \in D$
- [ ] Let $c \in R$. The function cf is defined on D by $(cf)(\tilde{x}) = \frac{1}{c} \times f(\tilde{x}), \tilde{x} \in D$
- [ ] If $m = 1$ and $g(\tilde{x}) \neq 0, \tilde{x} \in D$, then the function f/g is defined on D by $(f/g)(\tilde{x}) = f(\tilde{x})/g(\tilde{x}), \tilde{x} \in D$

No, the answer is incorrect.
Score: 0
Accepted Answers:
The sum function $f + g$ is defined on D by $(f + g)(\tilde{x}) = f(\tilde{x}) + g(\tilde{x}), \tilde{x} \in D$

If $m = 1$ and $g(\tilde{x}) \neq 0, \tilde{x} \in D$, then the function f/g is defined on D by $(f/g)(\tilde{x}) = f(\tilde{x})/g(\tilde{x}), \tilde{x} \in D$

1 point

Suppose $f : D \longrightarrow R$ is a function defined on a domain $D \subset R$. Which of the following can be f?

- [ ] $f(x) = 4 \pm \sqrt{x - 10}$
- [ ] $f(x) = \frac{x^2}{x+1}$
- [ ] $f(x) = log(\sqrt{x^4 + 3})$
- [ ] $f(x) = x^3 + x^2 - x$

No, the answer is incorrect.
Score: 0
Accepted Answers:

$f(x) = \frac{x^2}{x+1}$

$f(x) = log(\sqrt{x^4 + 3})$

$f(x) = x^3 + x^2 - x$

1 point

Which of the following statements is(are) TRUE?

- [ ] The output of a scalar valued multivariable function is always a real number.
- [ ] The output of a vector valued multivariable function can be a real number.
- [ ] The output of a single variable vector valued function is always a vector.
- [ ] The output of a vector valued multivariable function can not be a real number.

No, the answer is incorrect.
Score: 0
Accepted Answers:

The output of a scalar valued multivariable function is always a real number.
The output of a single variable vector valued function is always a vector.
The output of a vector valued multivariable function can not be a real number.

1 point

Consider the function $S(t) = (\cos t, \sin t, 2t)$, where t is real. Choose the set of correct options.

- [ ] S is a single-variable vector-valued function
- [ ] S is a multi-variable vector-valued function
- [ ] The domain of S is R and the co-domain of S is R^3
- [ ] The domain of S is R^3 and the co-domain of S is R^3

No, the answer is incorrect.
Score: 0
Accepted Answers:
SS is a single-variable vector-valued function
The domain of SS is RR and the co-domain of SS is R^3 R3
How many of the following functions is(are) a vector valued multivariable function?

- $f(x, y) = \frac{1}{3\pi}e^{x^2 + y^2}$ · f(x, y) = 3π1ex2+y2
- $f(x, y) = (x^3, y^2)$· f(x, y) = (x3, y2)
- $f(x, y) = (x + y, x - y, 2xy)$· f(x, y) = (x + y, x − y, 2xy)
- $f(x, y, z) = x^2 + xyz - z^2$ · f(x, y, z) = x2 + xyz − z2
- $f(x) = (x + 1, x - 1)$· f(x) = (x + 1, x − 1)
- $f(x, y, z) = (e^{x+y+z}, 0, 0)$· f(x, y, z) = (ex+y+z, 0, 0)

No, the answer is incorrect.
Score: 0
Accepted Answers:
(Type: Numeric) 3
1 point
1 point
Suppose $g: D \longrightarrow R^3$ g : D ⟶ R3 is a function defined on a domain $D \subset R^2$ D ⊂ R2. Which of the following functions may be gg?

☐ $g(x, y) = 4x + 3y - 12xy$g(x, y) = 4x + 3y − 12xy

☐ $g(x, y) = (x^4 + 1, y^4 + 1, xy)$g(x, y) = (x4 + 1, y4 + 1, xy)

☐ $g(x, y, z) = (e^x - e^z, e^z - e^y)$g(x, y, z) = (ex − ez, ez − ey)

☐ $g(x, y) = (sin(x + y), cos(x - y))$g(x, y) = (sin(x + y), cos(x − y))

☐ $g(x, y) = (0, sin(e^x), cos(e^x))$g(x, y) = (0, sin(ex), cos(ex))

No, the answer is incorrect.
Score: 0
Accepted Answers:
$g(x, y) = (x^4 + 1, y^4 + 1, xy)$g(x, y) = (x4 + 1, y4 + 1, xy)
$g(x, y) = (0, sin(e^x), cos(e^x))$g(x, y) = (0, sin(ex), cos(ex))

Level 2:

1 point
Which of the following are correct with respect to $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{16}}$f(x, y) = 1 − 9x2 − 16y2

$\sqrt{}$

, where x, yx, y are real?

☐ Domain of ff is $D = \{(x, y) | \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$D = {(x, y)|9x2 + 16y2 ≤ 1}

☐

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \geq 1\}$

☐

Range of f is $[0, 1]$

☐

Range of f is $[0, 1)$

☐

Range of f is $[1, \infty)$

No, the answer is incorrect.
Score: 0
Accepted Answers:

Domain of f is $D = \{(x, y) \mid \frac{x^2}{9} + \frac{y^2}{16} \leq 1\}$

Range of f is $[0, 1]$

1 point

A function $F: R^2 \rightarrow R^3$ has the form $F(x, y) = (P(x, y), Q(x, y), R(x, y))$. Choose the correct option(s).

☐

F is a multi-variable vector-valued function

☐

F is a single-variable vector-valued function

☐

P, Q, R are single-variable vector-valued functions

☐

P, Q, R are multi-variable vector-valued functions

☐

P, Q, R are multi-variable scalar-valued functions

No, the answer is incorrect.
Score: 0
Accepted Answers:

F is a multi-variable vector-valued function

P, Q, R are multi-variable scalar-valued functions

1 point

Consider $f(x, y) = \sqrt{1 - \frac{x^2}{9} - \frac{y^2}{4}}$. Note: An ellipse is of the form $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ where a, b are the dimensions of the ellipse.

☐

Ellipses of increasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

Ellipses of decreasing size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

Ellipses of constant size represent the curves $\{(x, y) \mid f(x, y) = c\}$, where c increases from 0 towards 1

☐

The curves $\{(x, y) \mid f(x, y) = c\}$, where $c \in [0, 1)$, cannot be represented by an ellipse

☐

$f(x, y) = 1$f(x, y) = 1 is a point

No, the answer is incorrect.
Score: 0
Accepted Answers:

Ellipses of decreasing size represent the curves $\{(x, y) | f(x, y) = c\}$\{(x, y) | f(x, y) = c\}, where cc increases from 00 towards 11

$f(x, y) = 1$f(x, y) = 1 is a point

1 point

Consider a point source $S = (1, 2, 3)$S = (1, 2, 3) radiating energy. The intensity I at a given point $P = (x, y, z)$P = (x, y, z) in space is inversely related to the square of the distance dd between SS and PP : $I(x, y, z) = \frac{k}{d^2}$I(x, y, z) = d2k, where kk is a real positive constant. Choose the correct option(s).

- [ ] Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1$\{(x, y, z) | (x − 1)2 + (y − 2)2 + (z − 3)2 = 1\} \}
- [ ] Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}$\{(x, y, z) | (x − 1)2 + (y − 2)2 + (z − 3)2 = 2\}
- [ ] Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 1$\{(x, y, z) | x2 + y2 + z2 = 1\}
- [ ] Intensity I is constant on $\{(x, y, z) | x^2 + y^2 + z^2 = 2$\{(x, y, z) | x2 + y2 + z2 = 2 \}

No, the answer is incorrect.
Score: 0
Accepted Answers:

Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 1$\{(x, y, z) | (x − 1)2 + (y − 2)2 + (z − 3)2 = 1\} \}

Intensity I is constant on $\{(x, y, z) | (x-1)^2 + (y-2)^2 + (z-3)^2 = 2\}$\{(x, y, z) | (x − 1)2 + (y − 2)2 + (z − 3)2 = 2\}